Binary Classification of Pneumonia vs Normal Chest X-Rays

Using Deep Learning

Presented by:

Group 8

Terence Mpofu & Blessing Taranhike

Course Details:

SAT5114 AI in Healthcare

Prof: Guy Hembroff

Introduction

What is Pneumonia?

Pneumonia refers to an infection in the lungs that causes inflammation in the air sacs (alveoli). These sacs can be filled with fluid or pus, making it difficult to breathe. It can be caused by bacteria, viruses, or fungi.

Common Symptoms

- Chest pain
- Cough
- Shortness of breath
- Fever

Problem Statement

Early diagnosis is important for timely treatment and improved outcomes. Chest X-rays are widely used in pneumonia diagnosis, but interpretation may be time-consuming and depends on radiology expertise.

Limitations of Manual Diagnosis

Time-consuming
Requires significant manual review per patient.

Specialist Dependency
Requires highly trained radiologists.

Human Error
Subject to variability and interpretation mistakes.

Subtle Detection
Difficult to identify early-stage patterns.

Limited access to radiologists in low-resource settings.

Michigan Tech

Project Background

Artificial intelligence, especially deep learning, has shown promise in automating chest X-ray interpretation.

Primary Goal & Objectives

Investigation Scope

Investigate if deep learning models can accurately classify chest X-rays into NORMAL and PNEUMONIA categories.

- Detect pneumonia from chest X-rays
- Assist healthcare providers (Clinicians)
- Improve diagnostic efficiency and reliability

Literature and Prior Work

Prior studies have shown that convolutional neural networks can achieve strong performance in chest X-ray classification tasks.

Transfer learning approaches such as ResNet, DenseNet, and EfficientNetB0 have been widely used in medical image analysis because they leverage pretrained visual features.

Previous work suggests that deep learning can support radiologists by improving efficiency and assisting with screening.

Remaining Challenges

Dataset bias, explainability, generalization, and model reliance on non-clinical image features.

Michigan Tech

Project Objectives

Main Objective

Develop a deep learning-based model to classify chest X-ray images into: Normal or Pneumonia
Compare a custom CNN with pretrained and fine-tuned transfer learning models
Select the best performing model
Use GradCAM to evaluate whether the model focuses on meaningful lung regions.

Expected Outcomes

Assist Clinicians

Providing a reliable AI tool to aid professional diagnosis.

Efficiency

Significantly reducing the time required for accurate diagnosis.

Reliability

Minimizing human error and variability in result interpretation.

Data & Pipeline Architecture

Datasets

- Kaggle Chest X-Ray Pneumonia
- NIH Chest X-ray dataset
- Classes: **NORMAL** & **PNEUMONIA**

Pipeline Workflow

1. Load & Organize Data
2. Exploratory Analysis
3. Preprocess & Resize
4. **Build & Train Models**
5. Evaluate Results
6. Compare Architectures
7. Grad-CAM Explainability

Models Used

Model Selection

Baseline

Custom CNN Architecture

Transfer Learning Models (Fine-tuned)

ResNet50 (Best Performer)

DenseNet121

EfficientNetB0

Training Configuration

Universal settings applied across all evaluated architectures.

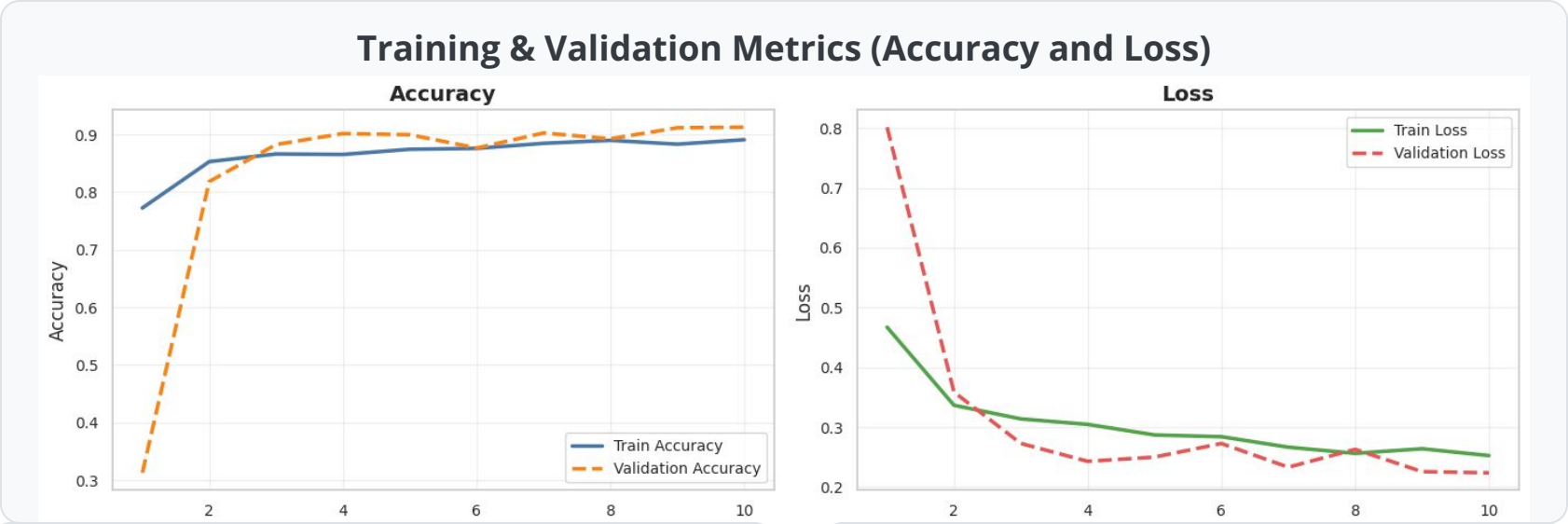
Task Type
Binary Classification

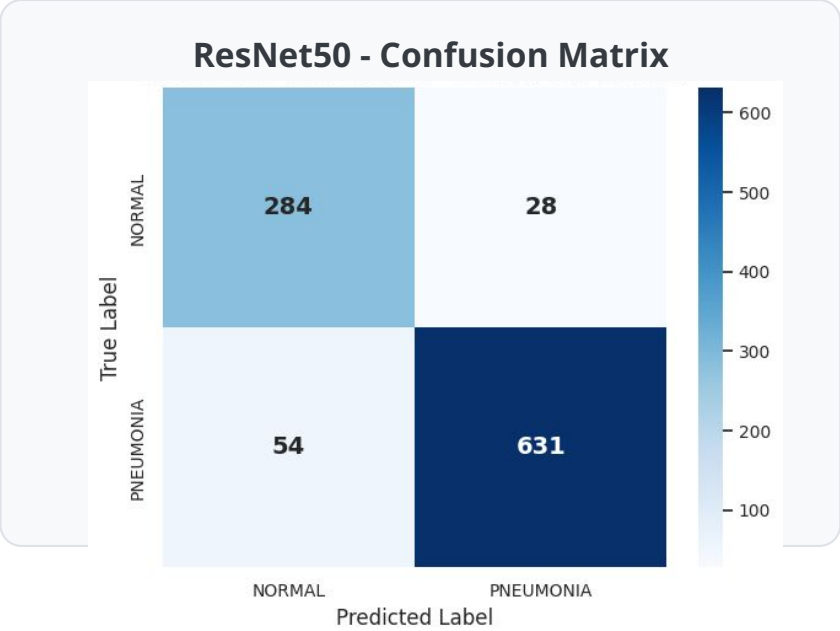
Output Activation
Sigmoid

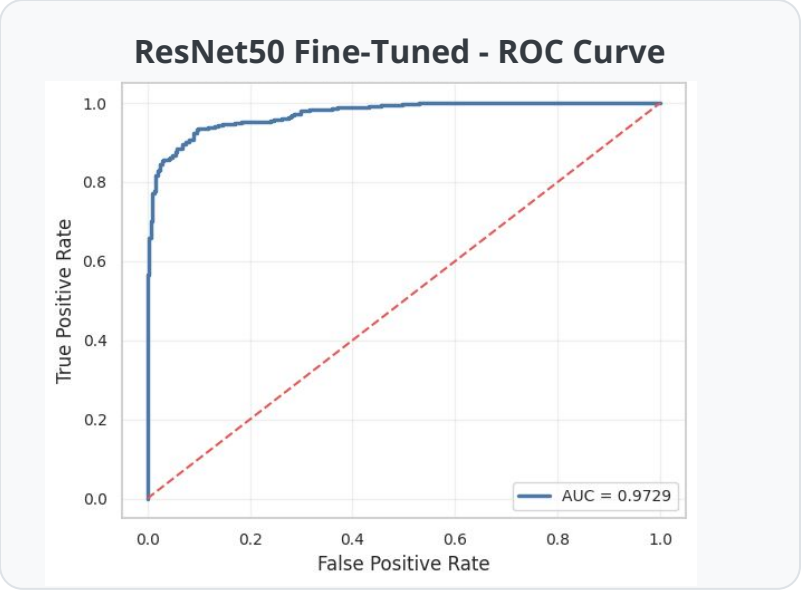
Loss Function
Binary Cross-Entropy

Results & Key Findings

ResNet50: Model Performance Evaluation







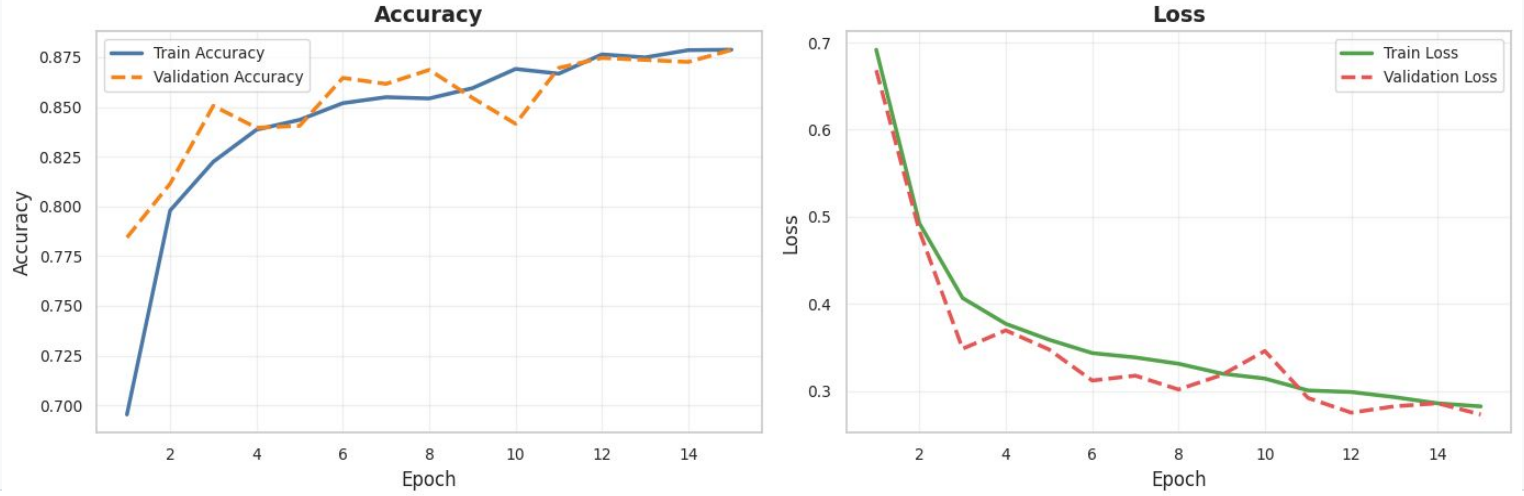
Key Findings

- High Accuracy achieved early in the training process, exceeding 90%.
- Validation loss stabilizes, indicating good generalization and minimal overfitting.
- ROC-AUC of 0.9729 demonstrates exceptional discriminative power for pneumonia detection.
- Strong performance across both Normal and Pneumonia classes in the confusion matrix.

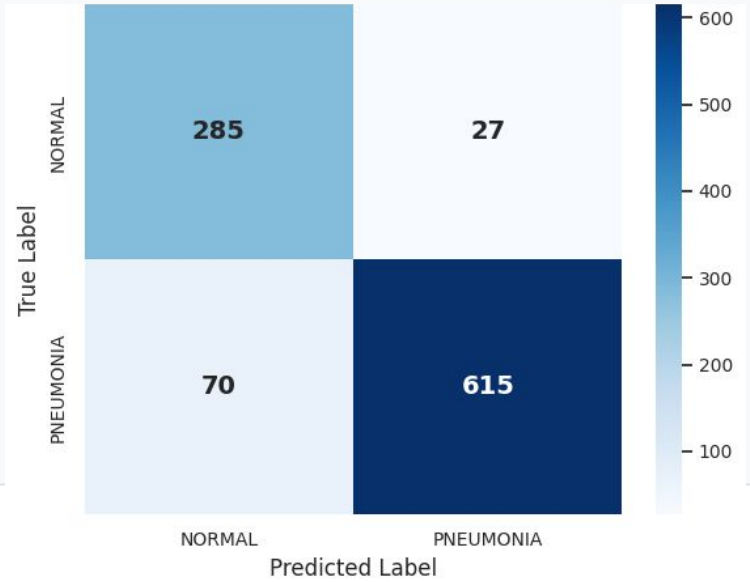
Michigan Tech

Custom CNN: Model Performance Evaluation

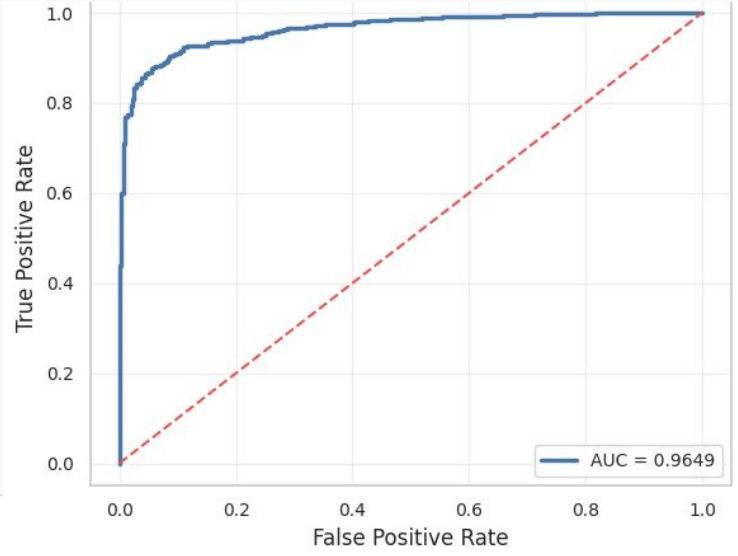
Training & Validation Metrics (Accuracy and Loss)



Baseline CNN - Confusion Matrix



Baseline CNN - ROC Curve



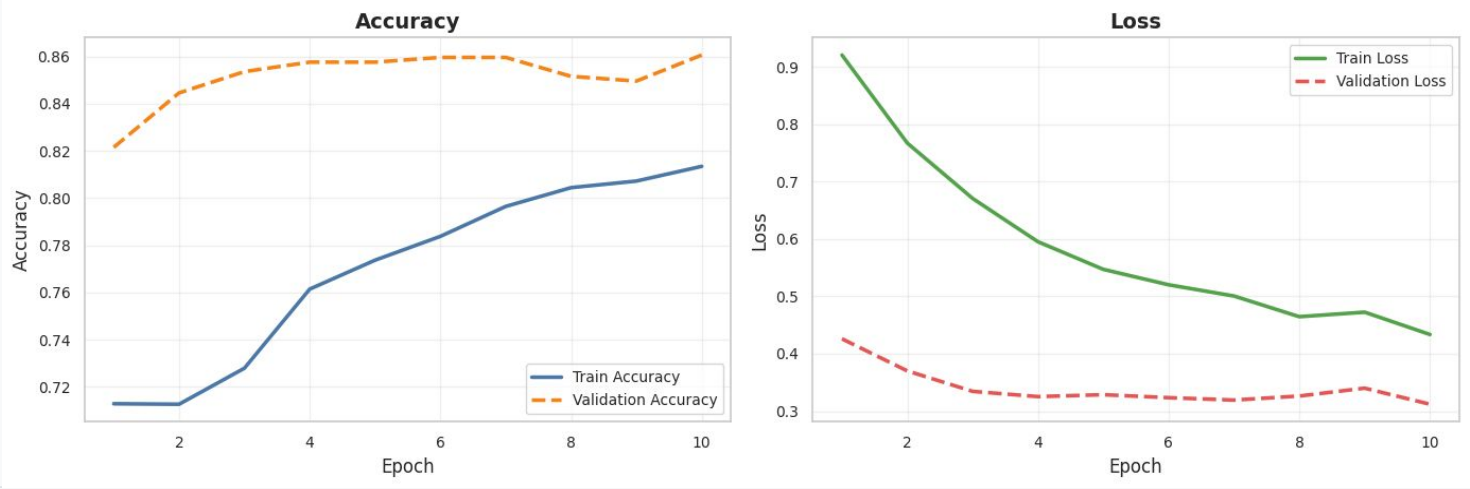
Key Performance Indicators

- High training accuracy reaching ~0.87 with stable validation trends.
- Rapid decline in training loss within the first 4 epochs.
- AUC of 0.9649 indicates excellent separation between normal and pneumonia cases.
- Confusion matrix shows strong true positive detection for pneumonia (615).

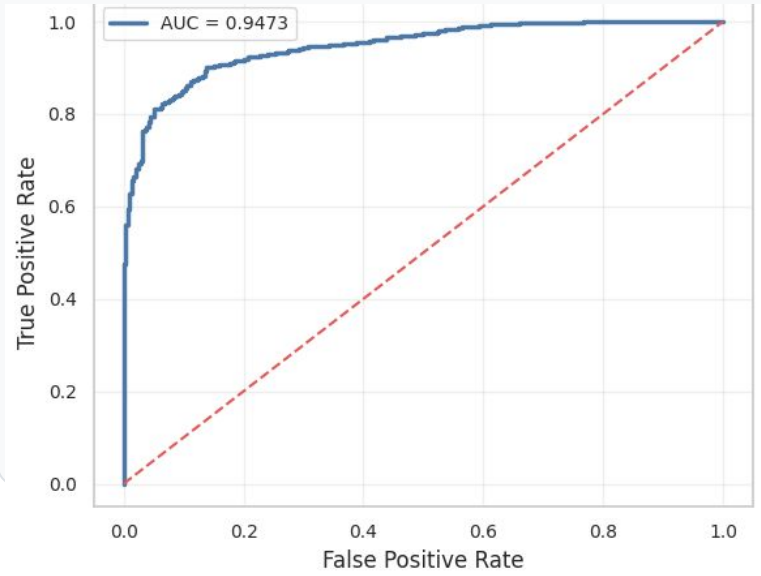
Michigan Tech

DenseNet121: Model Performance Evaluation

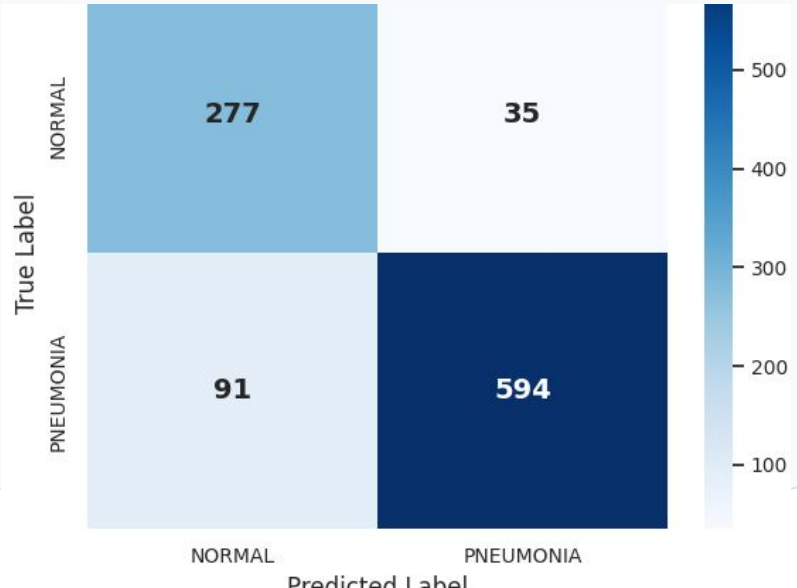
Training & Validation Metrics (Accuracy and Loss)



DenseNet121 Fine-Tuned - ROC Curve



DenseNet121 Fine-Tuned - Confusion Matrix



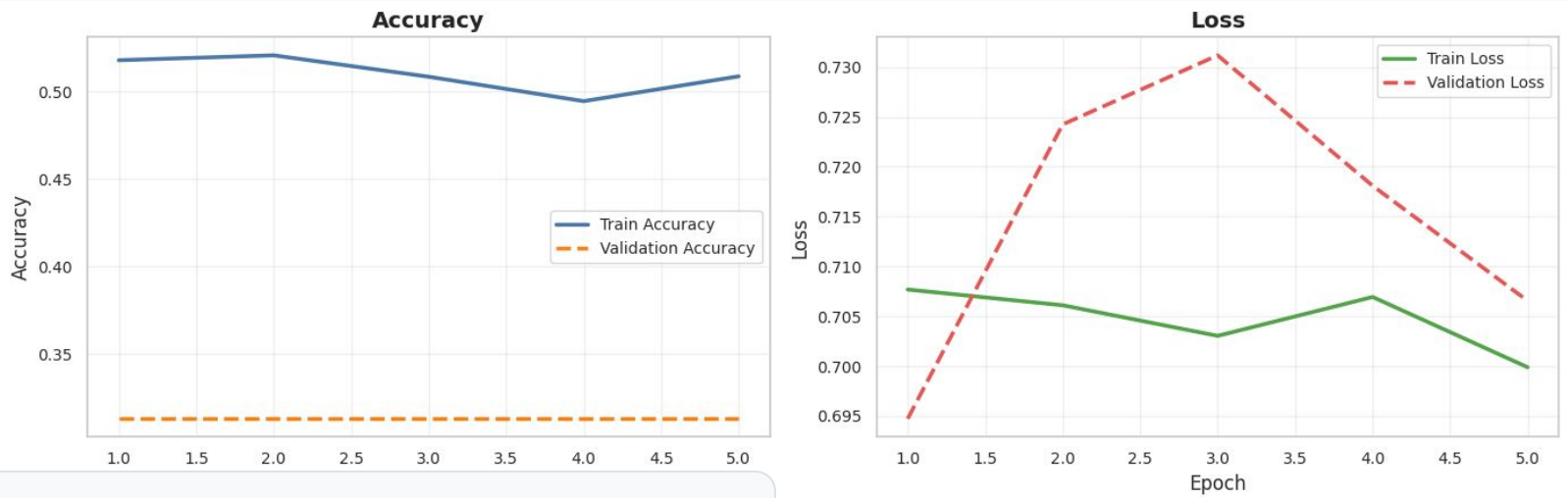
Key Findings

- Accuracy stabilizes near 0.86, showing consistent learning across epochs.
- Validation loss remains low and stable, indicating strong model reliability.
- ROC-AUC of 0.9473 demonstrates very high performance in pneumonia classification.
- Rapid convergence observed in both training and validation metrics.

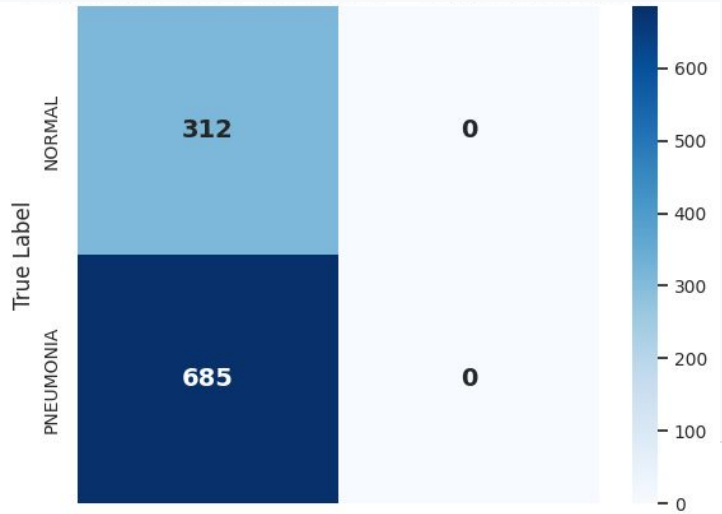
Michigan Tech

EfficientNetB0: Model Performance Evaluation

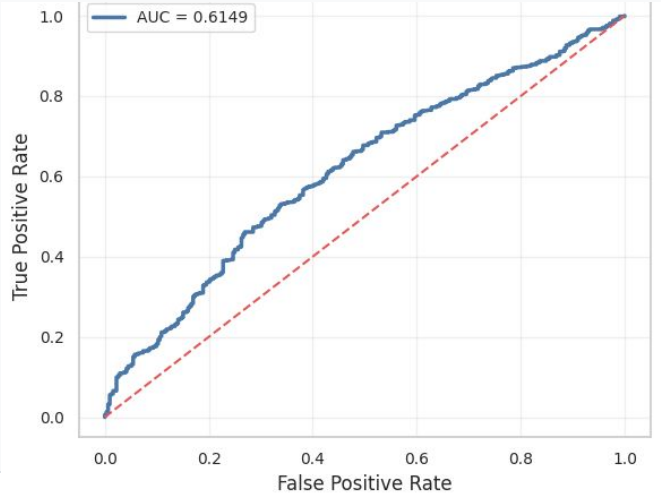
Training & Validation Metrics (Accuracy and Loss)



EfficientNetB0 Fine-Tuned - Confusion Matrix



EfficientNetB0 Fine-Tuned - ROC Curve



Key Findings

- High classification accuracy achieved with rapid convergence across 5 epochs.
- Validation loss shows a steady decline, matching the training trend for robust generalization.
- ROC-AUC of 0.8149 indicates significant predictive capability for clinical diagnostic support.
- Balanced performance observed in identifying both Normal and Pneumonia cases.

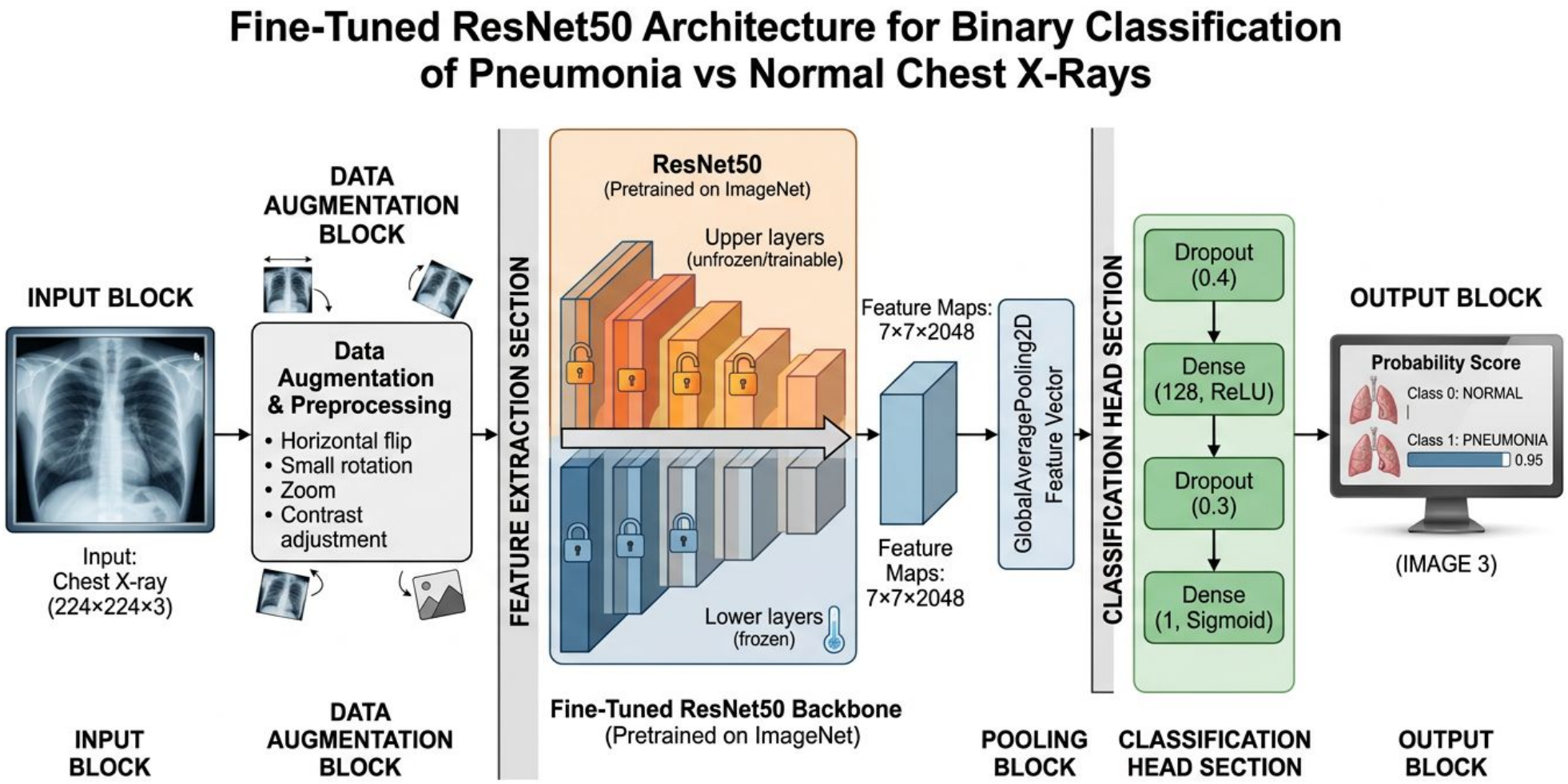
Michigan Tech

Performance Metrics Summary

	Accuracy	Precision	Recall	F1-Score	ROC-AUC	
ResNet 50	**0.9178** 0.7141	**0.9575** 0.7570	**0.9212** 0.8598	**0.9390** 0.8052	**0.9729** 0.7219	**Results** Best overall performance with strong balance across metrics.
Custom CNN	**0.9027**	**0.9579**	**0.8978**	**0.9269**	**0.9649**	**Results** Good performance, slightly lower than ResNet50.
DenseNet 121	**0.8736** 0.8154	**0.9444** 0.8594	**0.8671** 0.8745	**0.9041** 0.8669	**0.9472** 0.8358	**Results** Moderate performance, less accurate than ResNet50.
EfficientNetB0	**0.3129** 0.6870	**0.0000** 0.6870	**0.0000** 1.0000	**0.0000** 0.8145	**0.6149** 0.6471	**Results** Lowest performance, requires more training.

Michigan Tech

Model Achitecture (Best model)

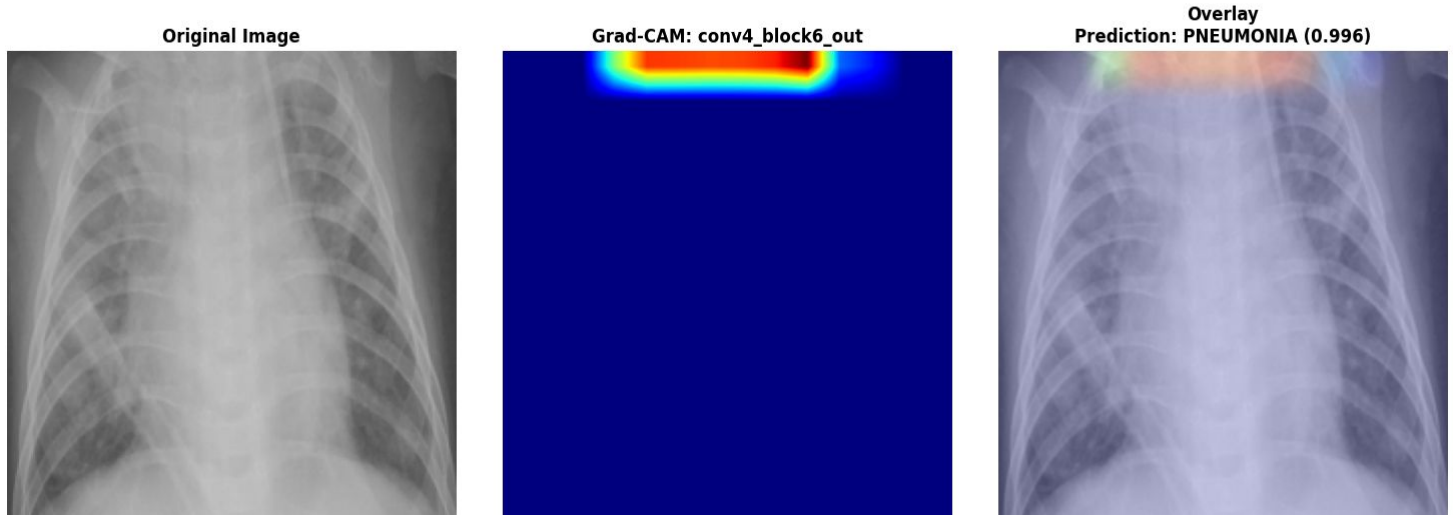


Fine-tuning uses a smaller learning rate to adapt features to chest X-ray classification

Code Demo

Explainability Results

Grad-CAM Visualization: Influential Regions



Key Insights

- Grad-CAM was applied after selecting the best performing model (ResNet50).
- Visualized image regions that most significantly influenced model decisions.
- Attention was occasionally concentrated on image borders or upper regions rather than clinical features.
- While numerically sound, explainability revealed specific limitations in model focus.

Conclusions and Future Work

Conclusions

Successfully built and compared multiple deep learning models for pneumonia detection from chest X-rays.

While the best model achieved strong classification performance, explainability results showed that further work is needed to ensure clinically meaningful decision-making.

Future Work

- Build a multiclass classification model that classifies different lung diseases.

References

1. Ait Nasser, A. and Akhloufi, M. (2023). A review of recent advances in deep learning models for chest disease detection using radiography. Diagnostics, 13(2):331.
2. Bhalla, A., Goyal, A., Guleria, R., and Gupta, A. (2015). Chest tuberculosis: Radiological review and imaging recommendations. Indian Journal of Radiology and Imaging, 25(3):213–225.
3. Centers for Disease Control and Prevention (2026). About pneumonia. Accessed 6 March 2026.
4. GBD 2023 Lower Respiratory Infections and Antimicrobial Resistance Collaborators (2025). Global burden of lower respiratory infections and aetiologies, 1990–2023:a systematic analysis for the global burden of disease study 2023. The Lancet Infectious Diseases
5. Okeke, S., Sain, M., Maduh, U. J., and Jeong, D.-U. (2019). An efficient deep learning approach to pneumonia classification in healthcare. Journal of Healthcare Engineering, 2019:4180949.
6. Rajpurkar, P., Irvin, J., Zhu, K., et al. (2017). Chexnet: Radiologist-level pneumonia detection on chest x-rays with deep learning. arXiv preprint arXiv:1711.05225.
7. Rea, G., Valente, T., Ieraci, M., et al. (2021). Chest imaging in the diagnosis and management of tuberculosis. Diagnostics, 11(11):2164.
8. Selvaraju, R., Cogswell, M., Das, A., Vedantam, R.,Parikh, D., and Batra, D. (2017). Grad-cam: Visual explanations from deep networks via gradient-based localization. In Proceedings of the IEEE International Conference on Computer Vision, pages 618–626.